



The Yerkes-Dodson Law (Yerkes & Dodson, 1908) or inverted U hypothesis suggests moderate arousal has the greatest positive effect on performance, with low or high arousal resulting in a deficit or loss of energy and a hindering of athletic ability (Miller, 1997; Wann & Church, 1998). This can be diagrammatically explained using an inverted U curve. Although this theory is most notably used to describe energy creation, maintenance and flow, it also applies to other psychological components such as self-confidence, which is later discussed. While the inverted U hypothesis links moderate arousal to optimal sporting outcomes, this middle level is more personally defined and explained by the zone of optimal functioning theory (Hanin, 1980, 1986, 1997).

Hanin's (1980, 1986, 1997) theory suggests individuals have different specific optimum arousal levels with optimum arousal occurring in an athlete's zone of optimal functioning. When in the "zone" athletes describe their movement as completely harmonious and effortless, with optimal performance achieved. Individual zones of optimal functioning are established through experience, practice, utilization of life and sporting memories as well as current and past physiological experiences. Once the "zone" is achieved, athletes should link this felt sense to some meaningful experiential anchor. This increases the accessibility of the "zone" during difficult sporting situations.

#### **2.3.1.1.3. Physiological arousal experience**

Depending on factors like expectancy, thoughts, attitude, motivation and memory, arousal may be linked to either positive or negative emotions. Low arousal is